

LOS Assignment 2

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1. Compare Permissions on Regular Files and Directories

Create a regular file named `sample.txt` and a directory named `sample_dir`. Create at least one file inside `sample_dir`.

Apply each of the following permissions using `chmod`. Perform the specified operations and record your observations.

Permission	Regular File (<code>sample.txt</code>)	Directory (<code>sample_dir</code>)
400 (r--)	The file can be read, but it cannot be modified because there is no write permission.	The directory can be listed, but it cannot be entered/accessed because there is no execute permission.
200 (-w-)	The file cannot normally be read, but its contents can be overwritten or changed if a command does not need to read the old contents.	The directory cannot be listed or entered, and a new file cannot be created because execute permission is missing.
100 (--x)	The file cannot be read. It can be executed only if it is actually an executable file/script and the system allows it to run	The directory cannot be listed, but it can be entered/accessed if you know the directory name and have the necessary path permissions.
500 (r-x)	The file can be read and executed, but it cannot be modified because there is no write permission.	The directory can be listed and entered, but files cannot be created or deleted because there is no write permission.
600 (rw-)	The file can be read and modified, but it cannot be executed.	The directory can be read/listed, but it cannot be entered or have files created because execute permission is missing.
700 (rwx)	The owner can read, modify, and execute the file. This gives full permission to the file.	The owner can list and enter the directory, create files, and delete files inside it.

```

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ ls
Lab2  sample_dir  sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ ls -l
total 8
drwxrwxr-x 4 sanjanarao sanjanarao 4096 Aug 20 22:15 Lab2
drwxrwxr-x 2 sanjanarao sanjanarao 4096 Aug 22 22:29 sample_dir
-rw-rw-r-- 1 sanjanarao sanjanarao    0 Aug 22 22:29 sample.txt

```

400 (r - -)

```

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ chmod 400 sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ chmod 400 sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ ls -l
total 8
drwxrwxr-x 4 sanjanarao sanjanarao 4096 Aug 20 22:15 Lab2
dr----- 2 sanjanarao sanjanarao 4096 Aug 22 22:29 sample_dir
-r----- 1 sanjanarao sanjanarao    0 Aug 22 22:29 sample.txt

```

Chmod 400 for sample.txt

```

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ cat sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ echo "Good day" > sample.txt
zsh: permission denied: sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
└─$ ./sample.txt
zsh: permission denied: ./sample.txt

```

So there is only permission to read, but no permission to write or execute.

Chmod 400 for sample_dir

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ touch sample_dir/file.txt
touch: cannot touch 'sample_dir/file.txt': Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cd sample_dir
cd: permission denied: sample_dir
```

So we can read the directory using ls but cannot write using touch and also cannot execute using cd.

200(-w-)

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 200 sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 200 sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls -l
total 8
drwxrwxr-x 4 sanjanarao sanjanarao 4096 Aug 20 22:15 Lab2
d-w----- 2 sanjanarao sanjanarao 4096 Aug 22 22:29 sample_dir
--w----- 1 sanjanarao sanjanarao    0 Aug 22 22:29 sample.txt
```

Chmod 200 sample.txt

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cat sample.txt
cat: sample.txt: Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ echo "Good day" > sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ./sample.txt
zsh: permission denied: ./sample.txt
```

So there is only permission to write, but no permission to read and execute

Chmod 200 sample_dir

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ touch sample_dir/file.txt
touch: cannot touch 'sample_dir/file.txt': Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cd sample_dir
cd: permission denied: sample_dir
```

Here we cannot read and execute, but we cannot read here as well because we can't navigate inside to create a file.

100(--x)

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 100 sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 100 sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls -ld sample.txt sample_dir
d--x----- 2 sanjanarao sanjanarao 4096 Aug 22 22:29 sample_dir
--x----- 1 sanjanarao sanjanarao    9 Aug 23 12:41 sample.txt
```

Chmod 100 sample.txt

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cat sample.txt
cat: sample.txt: Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ echo "Good day" > sample.txt
zsh: permission denied: sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ./sample.txt
zsh: permission denied: ./sample.txt
```

So there is no permission to read and write, but cannot execute as well because cannot execute a text script without reading it first.

Chmod 100 sample_dir

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls sample_dir
ls: cannot open directory 'sample_dir': Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ touch sample_dir/file.txt
touch: cannot touch 'sample_dir/file.txt': Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cd sample_dir
```

So we can execute the directory with cd, but cannot read with ls or write with touch.

500(r-x)

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 500 sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 500 sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls -ld sample.txt sample_dir
dr-x----- 2 sanjanarao sanjanarao 4096 Aug 22 22:29 sample_dir
-r-x----- 1 sanjanarao sanjanarao   9 Aug 23 12:41 sample.txt
```

Chmod 500 sample.txt

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cat sample.txt
Good day

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ echo "Good day" > sample.txt
zsh: permission denied: sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ./sample.txt
./sample.txt: 1: Good: not found
```

So there is only permission to read and execute, but no permission to write.

Chmod 500 sample_dir

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ touch sample_dir/file.txt
touch: cannot touch 'sample_dir/file.txt': Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cd sample_dir
```

So we can read the directory using ls but cannot write using touch and also can execute using cd.

600(rw-)

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 600 sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 600 sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls -ld sample.txt sample_dir
drw----- 2 sanjanarao sanjanarao 4096 Aug 22 22:29 sample_dir
-rw----- 1 sanjanarao sanjanarao   9 Aug 23 12:41 sample.txt
```

Chmod 600 sample.txt

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cat sample.txt
Good day

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ echo "Good day" > sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ./sample.txt
zsh: permission denied: ./sample.txt
```

So there is only permission to read and write , but no permission to execute.

Chmod 600 sample_dir

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ touch sample_dir/file.txt
touch: cannot touch 'sample_dir/file.txt': Permission denied

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cd sample_dir
cd: permission denied: sample_dir
```

So we can read the directory but cannot execute. Here, there should be an option to write, but as we cannot enter the directory, we cannot create a file inside it with touch.

700(rwx)

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 700 sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ chmod 700 sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls -ld sample.txt sample_dir
drwx----- 2 sanjanarao sanjanarao 4096 Aug 22 22:29 sample_dir
-rwx----- 1 sanjanarao sanjanarao   9 Aug 23 13:22 sample.txt
```

Chmod 700 sample.txt

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cat sample.txt
Good day

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ echo "Good day" > sample.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ./sample.txt
./sample.txt: 1: Good: not found
```

So we can read, write, execute the file.

Chmod 700 sample_dir

```
(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ ls sample_dir

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ touch sample_dir/file.txt

(sanjanarao@kali)-[~/LOS/LOS_Lab]
$ cd sample_dir
```

So we can read the directory using ls, write using touch, and also execute using cd.

Questions and Answers

1. What does read (r) permission allow you to do with a regular file?

Read permission allows you to view the contents of the file. For example, you can use cat sample.txt to see what is inside it.

2. What does read (r) permission allow you to do with a directory?

For a directory, read permission allows you to see the names of the files and subdirectories inside it. In other words, it allows you to list the directory contents.

3. What does write (w) permission mean for a regular file?

Write permission allows you to change the contents of the file. You can add, remove, or replace data in the file.

4. What does write (w) permission mean for a directory?

Write permission on a directory allows you to add, remove, or rename entries inside the directory. However, in practice, you also need execute (x) permission on the directory to actually perform these operations.

5. What does execute (x) permission mean for a regular file?

For a regular file, execute permission means the operating system allows the file to be run as a program or script, assuming its contents are something executable.

6. Why is execute (x) permission important for accessing a directory?

Execute permission on a directory means you are allowed to enter the directory and access items inside it. Without it, you may be able to see the names of files with read permission, but you generally cannot actually access those files through the directory.

7. Which permission combination allows the owner to fully access a regular file?

700 (rwx) gives the owner full permissions: read, write, and execute.

8. Which permission combination allows the owner to fully access a directory?

700 (rwx) gives the owner full permissions to list the directory, enter it, create files, delete files, and manage its contents.

9. Based on your observations, explain why the same r, w, and x permissions behave differently for a regular file and a directory.

The main reason is that a file and a directory are different types of objects. For a regular file, the permissions apply directly to the file's contents: r means reading the contents, w means changing the contents, and x means running the file.